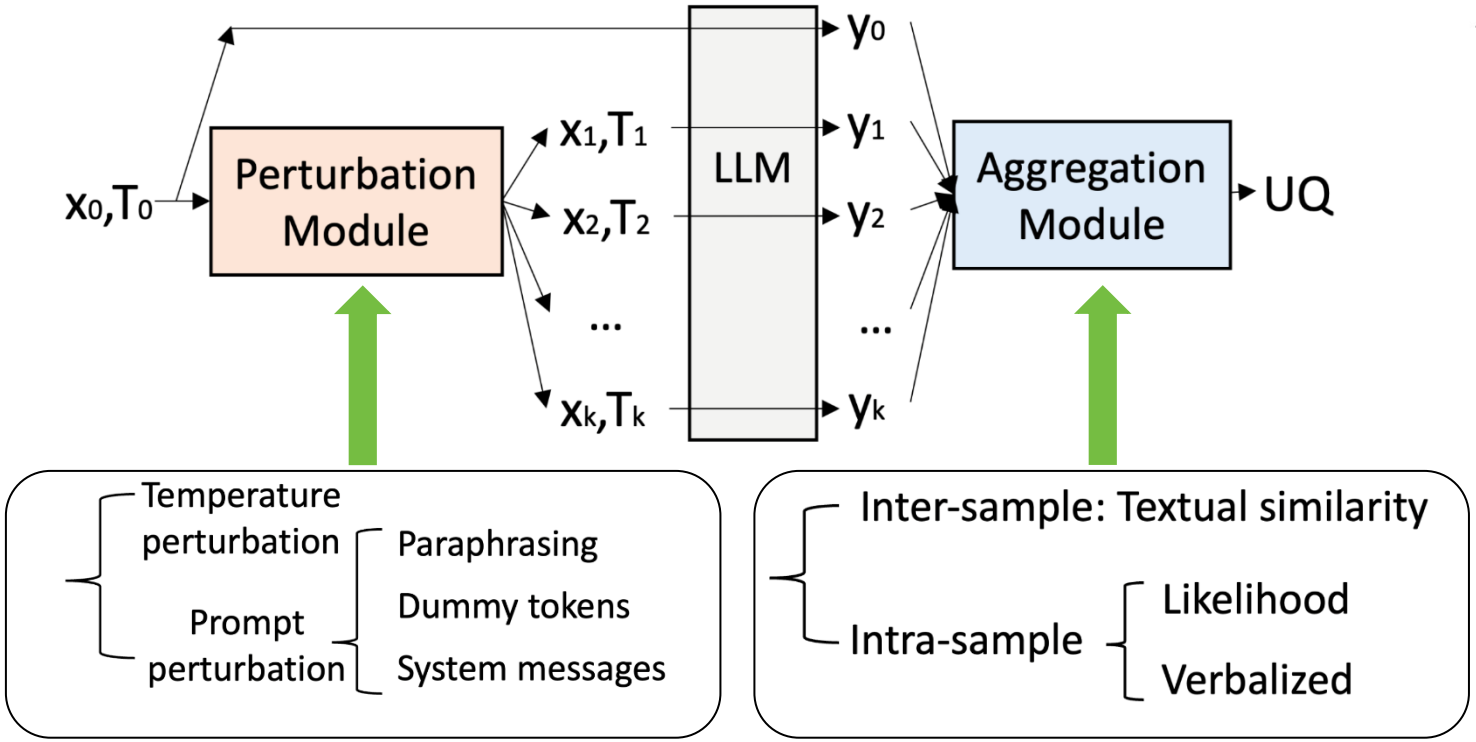
SPUQ

SPUQ[35]



Algorithm 1 SPUQ

Require: Original temperature T_0, original prompt x_0, perturbation and aggregation module hyperparameters (Fig. 2)

Ensure: Confidence score c

1: Obtain k perturbed inputs (T_i, x_i) from the perturbation module, where $i = 1$ to k.
2: **for** $j = 0$ to k **do**
3: Send (T_j, x_j) to LLM being tested
4: Obtain output y_j from LLM
5: **end for**
6: Send set $\{y_j\}$ to the aggregation module
7: Obtain the confidence score c
8: **return** c

- **Core Idea:**
  - accurately reflect the uncertainty of LLMs by perturbing the inputs and sampling the outputs.

Input x_i	**Prediction** y_i	**Likelihood**
Original		
Will Jay-Z reach the age of 60 before Kendrick Lamar?	No *(Incorrect)*	92%
Perturbed prompt		
Is it likely that Jay-Z will turn 60 before Kendrick Lamar does?	Yes	83%
Will Jay-Z hit the age of 60 before Kendrick Lamar does?	No	97%
Before Kendrick Lamar, will Jay-Z reach the age of 60?	Yes	94%
Is it possible that Jay-Z will turn 60 before Kendrick Lamar?	No	95%

[35] SPUQ: Perturbation-Based Uncertainty Quantification for Large Language Models, EACL 2024

SPUQ[35]

- **Task Type:** Open-ended generation

- **Introduction:** Sampling with perturbation for UQ (SPUQ) estimates uncertainty by perturbing the input and sampling multiple outputs. It entails generating a set of perturbations for LLM inputs, sampling outputs for each perturbation, and incorporating an aggregation module to unify these signals into a generalizable uncertainty estimate.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Consensus-based	Input/Decoding	Black-box	Post-hoc	Low	Calibration

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